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DLA - IS
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USAF - 99

USER SYMBOLS:

DD FORM 1426 672-1 (Coordinated)

P.A. NAVY - AS
Other Cust
USAF - 11
ARMY - AV

PROCUREMENT SPECIFICATION
MIL-B-23964

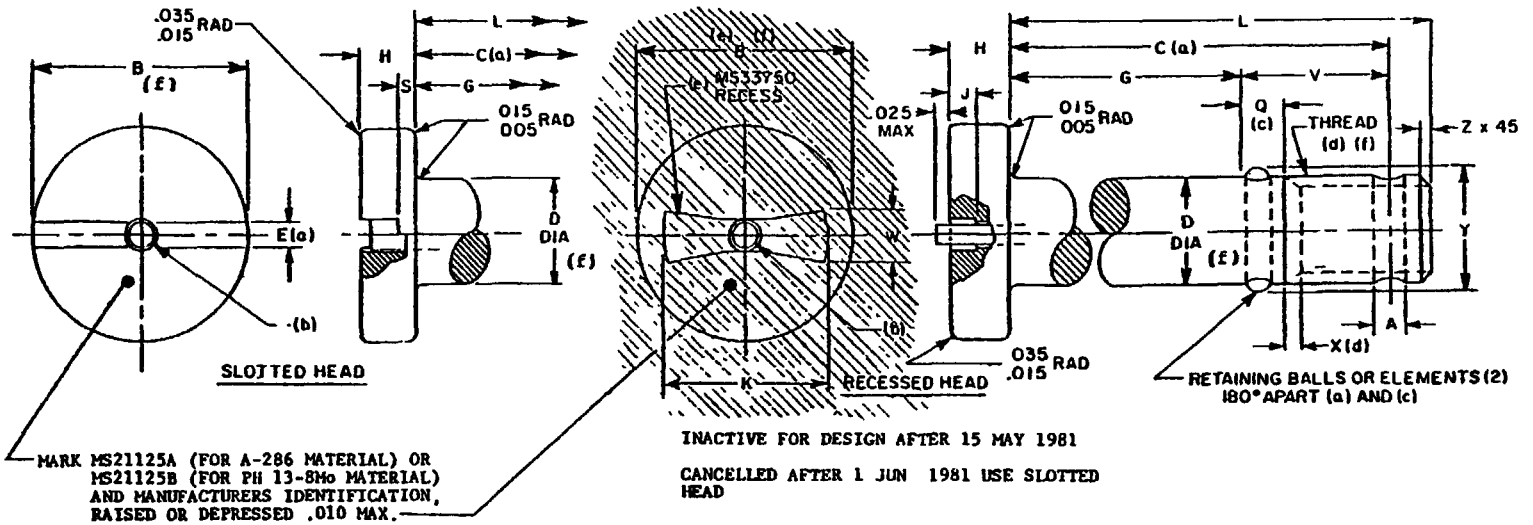
SUPERSEDES
MS21125 (ASG)

TITLE
BOLT, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Fcu, PAN HEAD, 450°F & 650°F

MILITARY STANDARD

MS21125

SHEET 1 OF 6



INACTIVE FOR DESIGN AFTER 15 MAY 1981

CANCELLED AFTER 1 JUN 1981 USE SLOTTED HEAD

(F) DENOTES CHANGE

FED SUP CLASS 5306

APPROVED 15 DEC 67 REVISED (D) 12 OCT 79 (E) 11 MARCH 1981 (F) 30 APRIL 1983

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PROJECT NO. 5306-0802

PLATE NO. 22065

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DD FORM 1 MAR 72 672-1 (Coordinated)

PROCUREMENT SPECIFICATION: MIL-B-23964

P.A.: NAVY - AS; Other Cust - USAF - 11, ARMY - AV

SUPERSEDES: MS21125(ASG)

TITLE: BOLT, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Ftu, .63 KSI Fsu, PAN HEAD, 450°F & 650°F

MILITARY STANDARD

MS21125

SHEET 2 OF 6

FED. SUP CLASS 5306

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PLATE NO. 22005

TABLE I

DIA DASH NO.	THREAD DESIGNATION MIL-S-8879	A MAX	A MIN	B (f) MAX	B (f) MIN	D (f) MAX	D (f) MIN	E MAX	E MIN	H MAX	H MIN	J MAX	J MIN	K MAX	(c) Q MAX	Q MIN	S MAX	S MIN	V BASIC	W MAX	W MIN	Y MAX	Y MIN	Z MAX	Z MIN	(e) RECESS NO.
3	.1900-32 UNJF-3A	.080	.070	.374	.360	1894	1889	057	047	.110	.095	.057	.054	.270	094	079	.032	.022	.293	.098	.094	.225	.215	041	.021	2
4	.2500-28 UNJF-3A	.086	076	.492	477	2492	2487	.073	063	141	.126	072	.069	.380	.125	.110			.333	.111	.107	.294	.284			3
5	.3125-24 UNJF-3A			625	.610	3117	3112	088	.078	.172	.157	.097	094	.545	.156	.141	.047	.037	375	.147	.143	.380	.370			4
6	.3750-24 UNJF-3A	.116	106	.740	.722	.3742	3737	.104	094	204	.189	111	.107	.630	.171	156			421	.164	.160	445	435			5
7	.4375-20 UNJF-3A			813	.795	4367	.4362			.235	.220	.118	.114	.730	.202	187	063	053	.460	.193	.188	514	504	057	.037	6
8	.5000-20 UNJF-3A			.915	.905	.4991	4986	.119	.109	.266	.251	.134	130	.815	.231	219	078	068	523	.221	.216	.599	.589			7
9	.5625-18 UNJF-3A	151	.141	1 035	1 015	5616	.5611			.297	.282	.149	.145	.900	.265	250	094	084	628	.262	.257	671	661	072	.052	8

GENERAL DIMENSIONAL AND MECHANICAL REQUIREMENTS

(a) COTTER PIN HOLE SHALL BE WITHIN .010 INCH OF BOLT CENTERLINE FOR BOLTS .312 DIA. AND SMALLER, AND WITHIN .015 INCH FOR BOLTS 375 DIA AND LARGER. THE PIN HOLE SHALL BE PERPENDICULAR TO BOLT CENTERLINE WITHIN 2°. LOCATION OF COTTER PIN HOLE FROM TOP OF HEAD IS GIVEN BY C-(G MIN +V) ±.010. THE COTTER PIN HOLE SHALL BE PERPENDICULAR TO THE SLOT IN HEAD WITHIN 5°. ORIENTATION OF BALLS OR ELEMENT TO COTTER PIN HOLE SHALL BE IN LINE WITHIN 3°. BALL OR ELEMENT PROTRUSION FROM THE BOLT CENTERLINE SHALL BE EQUAL OR LESS THAN 1/2 Y MAX.

(b) THE RETAINING DEVICE SHALL BE RELEASED ONLY BY DEPRESSING THE PLUNGER LOCATED IN THE HEAD OF THE BOLT. MAX DIAMETER OF PLUNGER FOR RECESS HEAD SHALL BE THE MIN VALUE FOR "M" DIMENSION ON MS33750.

(c) BALLS OR RETAINING ELEMENTS SHALL BE LOCATED WITHIN THE LIMITS OF "Q" DIMENSION, AND AT MAXIMUM PROTRUSION SHALL FIT WITHIN THE GROVES OF THE APPROPRIATE SPACER OF MS21126 AND MS21128 WITHOUT INTERFERENCE OR BINDING

(d) THREAD: IN ACCORDANCE WITH MIL-S-8879 EXCEPT THAT INCOMPLETE THREAD LENGTH (A) SHALL BE A MAXIMUM OF 1-1/2 AND A MINIMUM OF 1/2 THREAD PITCHES IN LENGTH AND MAJOR DIA SHALL BE .001 LESS THAN THE MEASURED SHANK DIA

(e) RECESS SHALL BE LOCATED AT TRUE POSITION "B" WITHIN .010 AND SHALL CONFORM TO MS33750 EXCEPT THAT RECESS DIMPLE IS OPTIONAL.

(f) CONCENTRICITY. D AND B TO BE CONCENTRIC TO P.D. OF THREAD WITHIN .002 TIR AT MAXIMUM MATERIAL CONDITION.

APPROVED 15 DEC 67 REVISED (F) FOR CHANGES SEE SHEET 6

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USER SYMBOLS:

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Other Cust
USAF - 11
ARMY - AV

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 83 KSI Ftu, PAN HEAD, 450°F & 650°F

PROCUREMENT SPECIFICATION
MIL-B-23964

SUPERSEDES
MS 21125 (ASC)

MILITARY STANDARD
MS21125

SHEET 3 OF 6

DD FORM 1 JAN 72 672-1 (Coordinated)
PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.
PLATE NO. 15320

FED. SUP CLASS 5306

TABLE II

GRIP DASH NO.	NOMINAL SIZE - THREAD DESIGNATION G-GRIP LENGTH		1900 (10) 32 UNJF-3A L-LENGTH		.2500 (1/4) 28 UNJF-3A L-LENGTH		.3125 (5/16) 24 UNJF-3A L-LENGTH		3750 (3/8) 24 UNJF-3A L-LENGTH		.4375 (7/16) 20 UNJF-3A L-LENGTH		.5000 (1/2) 20 UNJF-3A L-LENGTH		.5625 (9/16) 18 UNJF-3A L-LENGTH		GRIP DASH NO.
	MAX.	MIN.	MAX.	MIN.	MAX.	MIN	MAX	MIN.	MAX.	MIN.	MAX.	MIN.	MAX.	MIN	MAX.	MIN.	
03	.204	.188	.580	.560	.626	.606											03
04	.266	.250	642	.622	.688	.668	.748	.728	.809	.789							04
05	.328	312	.705	.685	.751	.731	.810	.790	871	.851							05
06	.390	.375	767	.747	.813	.793	.873	.853	934	.914	.978	.958	1.041	1.021			06
07	.454	438	.830	.810	.876	.856	.935	.915	996	.976	1.040	1 020	1.103	1.083	1.244	1 224	07
08	.516	500	.892	.872	938	.918	.998	.978	1.059	1.039	1.103	1 083	1.166	1.146	1.306	1.286	08
09	.578	562	.955	.935	1.001	.981	1.060	1.040	1.121	1.101	1.165	1.145	1.228	1.208	1.369	1.349	09
10	641	.625	1.017	.997	1.063	1.043	1.123	1.103	1.184	1.164	1.228	1.208	1.291	1.271	1.431	1 411	10
11	.704	688	1 080	1.060	1.126	1.106	1.185	1.165	1.246	1.226	1.290	1.270	1.353	1.333	1.494	1 474	11
12	.766	.750	1.142	1.122	1.188	1.168	1.248	1.228	1 309	1.289	1.353	1 333	1.416	1.396	1.556	1.536	12
13	.828	812	1.205	1 185	1.251	1.231	1.310	1.290	1.371	1.351	1.415	1 395	1.478	1.458	1.619	1.599	13
14	891	875	1.267	1.247	1 313	1.293	1.373	1 353	1.434	1 414	1.478	1.458	1.541	1.521	1.681	1 661	14
15	954	938	1.330	1.310	1.376	1.356	1.435	1.415	1.496	1.476	1.540	1.520	1.603	1.583	1.744	1.724	15
16	1.016	1 000	1.392	1.372	1.438	1 418	1.498	1.478	1 559	1.539	1.603	1.583	1.666	1.646	1.806	1.786	16
17	1 078	1.062	1.455	1.435	1.501	1.481	1.560	1.540	1 621	1.601	1.665	1 645	1.728	1.708	1.869	1.849	17
18	1.141	1 125	1.517	1.497	1.563	1.543	1 623	1.603	1.684	1.664	1.728	1.708	1.791	1.771	1.931	1.911	18
19	1.203	1.188	1 580	1.560	1 626	1.606	1.685	1.665	1.746	1.726	1.790	1.770	1.853	1.833	1.994	1.974	19
20	1 266	1 250	1 642	1.622	1.688	1.668	1.748	1.728	1.809	1.789	1.853	1 833	1.916	1.896	2.056	2.036	20
21	1.328	1.312	1.705	1.685	1.751	1.731	1.810	1.790	1.871	1.851	1.915	1 895	1 978	1.958	2.119	2.099	21
22	1.391	1.375	1 767	1.747	1.813	1.793	1.873	1.853	1.934	1.914	1.978	1.958	2.041	2 021	2.181	2.161	22
23	1.454	1 438	1.830	1.810	1.876	1.856	1.935	1.915	1 996	1.976	2.040	2.020	2.103	2 083	2.244	2.224	23
24	1 516	1 500	1.892	1.872	1.938	1.918	1.998	1.978	2.059	2.039	2 103	2.083	2.166	2.146	2.306	2.286	24
25	1 578	1.562	1 954	1 934	2.001	1.981	2.060	2.040	2.121	2.101	2.165	2.145	2.228	2.208	2.369	2.349	25
26	1.641	1 625	2 017	1 997	2.063	2 043	2 123	2.103	2 184	2.164	2.228	2.208	2.291	2.271	2.431	2 411	26
27	1.704	1 688	2.080	2.060	2.126	2.106	2.185	2.165	2 246	2.226	2.290	2 270	2.353	2 333	2.494	2.474	27
28	1.766	1 750	2 142	2 122	2 188	2.168	2.248	2.228	2.309	2 289	2 353	2.333	2.416	2.396	2.556	2.536	28
29	1.828	1.812	2.205	2.185	2.251	2.231	2.310	2.290	2.371	2.351	2.415	2 395	2.478	2.458	2.619	2.599	29
30	1.891	1.875	2 267	2.247	2.313	2.293	2.373	2.353	2.434	2 414	2.478	2.458	2.541	2 521	2.681	2.661	30
31	1.953	1.937	2 330	2.310	2.376	2.356	2.435	2.415	2.496	2.476	2.540	2.520	2.603	2.583	2 744	2 724	31
32	2.016	2.000	2.392	2.372	2.438	2.418	2.498	2.478	2.559	2.539	2.603	2.583	2.666	2.646	2 806	2.786	32
33	2.078	2.062	2.455	2.435	2.501	2.481	2.560	2.540	2 621	2.601	2.665	2.645	2.728	2 708	2.869	2 849	33
34	2 .141	2.125	2.517	2 497	2.563	2.543	2 623	2 603	2.684	2 664	2.728	2 708	2.791	2.771	2.931	2.911	34

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NAVY - AS
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PLATE NO 17788

P.A. NAVY - AS
Other Cust
— USAF-11
ARMY-AV

PROCUREMENT SPECIFICATION
MIL-B-23864

SUPERSEDES
MS 21125 (ASG)

TITLE
BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES
90 KSI Fsu, 60 KSI Ftu, PAN HEAD, 450°F & 950°F

SHEET 4 OF 6

MS21125

MILITARY STANDARD

FED. SUP CLASS
5306

TABLE III

GRIP DASH NO.	NOMINAL SIZE		1900 (10)		.2500 (1/4)		.3125 (5/16)		.3750 (3/8)		.4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO.
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
35	2.204	2.188	2.580	2.560	2.626	2.606	2 685	2 665	2.746	2.726	2 790	2.770	2.853	2 833	2.994	2.974	35
36	2.266	2 250	2.642	2.622	2.688	2 668	2.748	2.728	2.809	2.789	2 853	2.833	2.916	2.896	3.056	3.036	36
37	2.328	2.312	2.705	2 685	2 751	2 731	2.810	2.790	2.871	2.851	2.915	2.895	2.978	2.958	3.119	3.099	37
38	2.390	2.375	2 767	2 747	2.813	2.793	2.873	2 853	2.934	2.914	2.978	2.958	3 041	3.021	3.181	3.161	38
39	2.454	2.438	2.830	2 810	2.876	2.856	2.935	2.915	2.996	2.976	3.040	3.020	3.103	3.083	3.244	3.224	39
40	2.516	2.500	2.892	2.872	2.938	2.918	2 998	2.978	3 059	3.039	3.103	3.083	3.166	3.146	3.306	3.286	40
41	2.578	2.562	2 954	2 934	3 001	2.981	3.060	3.040	3.121	3.101	3.165	3 145	3.228	3.208	3.369	3 349	41
42	2 641	2.625	3.017	2 997	3 063	3 043	3.123	3.103	3.184	3.164	3.228	3.208	3.291	3.271	3 431	3 411	42
43	2.704	2 688	3.080	3 060	3 126	3 106	3 185	3 165	3.246	3.226	3.290	3.270	3.353	3.333	3.494	3 474	43
44	2 766	2.750	3.142	3.122	3.188	3 168	3.248	3.228	3.309	3.289	3.353	3.333	3.416	3 396	3 556	3.536	44
45	2 828	2.812	3 205	3 185	3.251	3.231	3.310	3.290	3.371	3.351	3 415	3.395	3.478	3 458	3 619	3.599	45
46	2.891	2.875	3.267	3.247	3 313	3.293	3.373	3 353	3.434	3 414	3 478	3.458	3.541	3 521	3.681	3.661	46
47	2 954	2 938	3 330	3.310	3.376	3.356	3.435	3 415	3.496	3.476	3.540	3.520	3.603	3.583	3 744	3.724	47
48	3 016	3.000	3 392	3.372	3 438	3.418	3.498	3.478	3 559	3.539	3 603	3.583	3.666	3 646	3.806	3 786	48
49	3.078	3.062	3.455	3 435	3.501	3 481	3.560	3 540	3.621	3.601	3 665	3.645	3.728	3.708	3.869	3.849	49
50	3.141	3.125	3 517	3 497	3 563	3.543	3 623	3 603	3.684	3 664	3 728	3 708	3 791	3.771	3 931	3.911	50
51	3.203	3 188	3.580	3.560	3 626	3 606	3.685	3 665	3 746	3 726	3 790	3 770	3.853	3.833	3.994	3.974	51
52	3.266	3.250	3 642	3 622	3 688	3 668	3.748	3 728	3.809	3 789	3.853	3.833	3.916	3.896	4.056	4.036	52
53	3.328	3 312	3.705	3 685	3.751	3.731	3 810	3.790	3.871	3 851	3.915	3.895	3.978	3 958	4.119	4.099	53
54	3 391	3.375	3.767	3 747	3.813	3.793	3.873	3 853	3.934	3.914	3.978	3 958	4.041	4.021	4 181	4.161	54
55	3.454	3 438	3.830	3 810	3 876	3.856	3.935	3.915	3.996	3.976	4 040	4.020	4.103	4 083	4.244	4.224	55
56	3 516	3 500	3.892	3.872	3 938	3.918	3 998	3 978	4.059	4.039	4 103	4 083	4.166	4.146	4.306	4.286	56
57	3.578	3.562	3.954	3.934	4 001	3.981	4.060	4 040	4 121	4 101	4 165	4 145	4.228	4.208	4.369	4.349	57
58	3 641	3 625	4.017	3 997	4 063	4 043	4.123	4.103	4 184	4 164	4 228	4.208	4.291	4 271	4 431	4 411	58
59	3 704	3.688	4 080	4 060	4 126	4 106	4 185	4 165	4 246	4.226	4 290	4.270	4 353	4.333	4 494	4 474	59
60	3.766	3 750	4.142	4 122	4 188	4 168	4.248	4.228	4 309	4.289	4 353	4 333	4 416	4.396	4 556	4 536	60
61	3.828	3 812	4.205	4 185	4.251	4.231	4 310	4.290	4.371	4 351	4 415	4 395	4 478	4 458	4 619	4.599	61

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USAF-99

USER SYMBOLS:

TABLE IV

GRIP DASH NO	NOMINAL SIZE		1900 (10)		2500 (1/4)		3125 (5/16)		3750 (3/8)		4375 (7/16)		.5000 (1/2)		.5625 (9/16)		GRIP DASH NO
	THREAD DESIGNATION		32 UNJF-3A		28 UNJF-3A		24 UNJF-3A		24 UNJF-3A		20 UNJF-3A		20 UNJF-3A		18 UNJF-3A		
	G-GRIP LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		L-LENGTH		
	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN	
62	3 891	3.875	4.267	4.247	4 313	4 293	4.373	4 353	4 434	4 414	4 478	4.458	4.541	4 521	4 681	4.661	62
63	3 953	3 937	4 330	4 310	4 376	4 356	4 435	4 415	4.496	4.476	4.540	4.520	4.603	4.583	4 744	4 724	63
64	4 016	4.000	4 392	4 372	4 438	4.418	4 498	4 478	4 559	4.539	4 603	4.583	4.666	4.646	4.806	4.786	64
65	4.078	4.062	4 455	4 435	4 501	4 481	4.560	4 540	4 621	4 601	4.665	4 645	4 728	4 708	4 869	4.849	65
66	4.141	4 125	4 517	4 497	4.563	4.543	4 623	4 603	4.684	4 664	4.728	4.708	4.791	4 771	4.931	4.911	66
67	4.203	4 188	4 580	4.560	4 626	4 606	4.685	4 665	4 746	4 726	4 790	4 770	4 853	4.833	4 994	4 974	67
68	4.266	4.250	4 642	4.622	4 688	4 668	4 748	4 728	4 809	4 789	4 853	4.833	4.916	4.896	5 056	5 036	68
69	4 328	4.312	4 705	4 685	4 751	4.731	4.810	4 790	4 871	4.851	4 915	4.895	4.978	4 958	5 119	5.099	69
70	4 391	4 375	4 767	4 747	4 813	4.793	4.873	4.853	4 934	4 914	4 978	4 958	5.041	5.021	5 181	5 161	70
71	4 454	4 438	4 830	4.810	4 876	4.856	4 935	4 915	4 996	4 976	5.040	5 020	5.103	5 083	5.244	5.224	71
72	4.516	4.500	4 892	4 872	4 938	4 918	4 998	4 978	5 059	5.039	5.103	5 083	5.166	5 146	5 306	5.286	72
73	4.578	4.562	4 954	4.934	5 001	4 981	5 060	5 040	5.121	5 101	5.165	5 145	5.228	5.208	5 369	5.349	73
74	4 641	4.625	5.017	4.997	5 063	5 043	5.123	5 103	5 184	5.164	5.228	5.208	5.291	5 271	5 431	5 411	74
75	4 704	4 688	5.080	5 060	5 126	5 106	5.185	5 165	5 246	5.226	5 290	5.270	5.353	5 333	5 494	5 474	75
76	4 766	4.750	5.142	5 122	5 188	5.168	5.248	5.228	5 309	5.289	5 353	5.333	5.416	5.396	5.556	5.536	76
77	4.828	4.812	5.205	5 185	5.251	5.231	5 310	5.290	5 371	5.351	5 415	5.395	5.478	5.458	5 619	5 599	77
78	4.891	4.875	5.267	5.247	5 313	5.293	5.373	5 353	5 434	5 414	5 478	5 458	5 541	5.521	5 681	5.661	78
79	4 953	4.937	5.330	5.310	5.376	5.356	5.435	5.415	5.496	5.476	5 540	5.520	5 603	5.583	5.744	5.724	79
80	5.016	5.000	5 392	5 372	5 438	5.418	5 498	5 478	5.559	5.539	5.603	5.583	5.666	5.646	5 806	5.786	80
81	5.078	5 062	5.455	5.435	5 501	5.481	5.560	5.540	5.621	5.601	5.665	5.645	5.728	5 708	5 869	5 849	81
82	5 141	5 125	5 517	5.497	5 563	5.543	5.623	5.603	5 684	5.664	5 728	5.708	5.791	5.771	5.931	5.911	82
83	5.203	5.188	5.580	5.560	5 626	5.606	5.685	5 665	5 746	5.726	5 790	5 770	5 853	5.833	5.994	5.974	83
84	5.266	5.250	5 642	5 622	5 688	5.668	5.748	5.728	5.809	5.789	5.853	5.833	5.916	5.896	6.056	6 036	84
85	5.328	5 312	5 705	5.685	5 751	5.731	5.810	5.790	5.871	5.851	5.915	5 895	5.978	5 958	6.119	6 099	85
86	5.390	5 375	5.767	5.747	5 813	5 793	5.873	5.853	5.934	5 914	5.978	5.958	6.041	6.021	6.181	6.161	86

DD FORM 672-1 (Coordinated)

PROCUREMENT SPECIFICATION: MIL-B-23984

P.A. NAVY - AS
Other Cust
USAF-11
ARMY-AV

SUPERSEDES: MS 21125 (ASG)

TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Ftu, PAN HEAD, 450°F & 650°F

MILITARY STANDARD

MS21125

SHEET 5 OF 6

FED. SUP CLASS 5306

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO 17709

APPROVED 15 Dec 67

REVISED (F) For changes see sheet 6

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This military standard is approved for use by all Departments & Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

FED SUP CLASS 5306

TABLE V

SIZE	SHANK DOUBLE SHEAR ULTIMATE STRENGTH LBS MIN	BOLT AND NUT ASSEMBLY ULTIMATE TENSILE STRENGTH LBS MIN	RETAINING DEVICE ULTIMATE TENSILE LOAD LBS MIN
.1900	5,150	1,110	110
.2500	9,200	2,040	230
.3125	14,400	3,250	510
.3750	20,600	5,050	575
.4375	28,000	6,800	710
.5000	36,800	9,250	1,160
.5625	46,000	11,300	1,420

NOTES

1. MATERIALS, MAGNETIC. BOLT AND PLUNGER: CORROSION RESISTING STEEL PH 13-8Mo CHEMISTRY OF AMS5629. BALLS OR ELEMENT: CORROSION RESISTING STEEL PER QQ-S-763 CLASS FSC 4400. SPRING: CORROSION RESISTING STEEL WIRE, SPRING TEMPER PER AMS 5673, 17-7 PH OR QQ-W-423, COMPOSITION 302, CONDITION B

2. HEAT TREATMENT: BOLT AND PLUNGER: PER MIL-H-6875 TO CONDITION H1000. BALLS: PER MIL-H-6875 TO ROCKWELL C55 TO C62

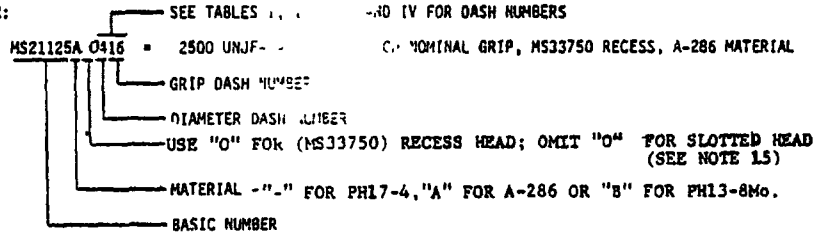
3. MATERIALS NONMAGNETIC: BOLT AND PLUNGER: CORROSION RESISTING STEEL A-286 CHEMISTRY OF AMS 5737. BALLS OR ELEMENT: CORROSION RESISTING STEEL PER MIL-S-7720. SPRING: NICKEL-CHROMIUM-IRON ALLOY PER QQ-W-390 CONDITION C OR NICKEL-ALLOY SPRING PER AMS 5699, SPRING TEMPER

4. PROTECTIVE TREATMENT: CRES MATERIAL PASSIVATED PER QQ-P-35

5. SURFACE TEXTURE. UNTHREADED SHANK AND BEARING SURFACE SHALL BE 32 MICROINCHES, ALL OTHER SURFACES SHALL BE 125 MICROINCHES IN ACCORDANCE WITH ASA B46.1.

6. BOLTS SHALL BE FREE FROM BURRS AND SLIVERS

(F) 7. EXAMPLE OF PART NUMBER:



8. DESIGN USAGE: FOR DESIGN AND USAGE LIMITATIONS SEE MS33602

9. MAGNETIC PERMEABILITY: FOR BOLT ASSEMBLY MADE FROM A-286 MATERERIAL MAGNETIC PERMEABILITY SHALL BE LESS THAN 2.0 (AIP=1.0) FOR A FIELD STRENGTH = 200 ORSTEADS USING MAGNETIC PERMEABILITY INDICATOR PER MIL-I-17214 OR EQUIVALENT.

10. TEMPERATURE LIMITATION: 450°F MAX FOR PH 13-Mo AND 650°F MAX. FOR A-286 MATERIAL.

(F) 11. INTERCHANGEABILITY RELATIONSHIP: BOLTS MADE FROM PH17-4 (CODE "-") ARE INACTIVE FOR NEW DESIGN AND CANCELLED AFTER 15 MAY 1981.

BOLTS MADE FROM CRES PH13-8Mo (CODE "B") MAY BE SUBSTITUTED FOR BOLTS MADE FROM PH17-4 IN ALL APPLICATIONS.

BOLTS MADE FROM PH17-4 (CODED "-") SHALL NOT BE USED IN PLACE OF BOLTS MADE FROM CRES PH13-8Mo.

BOLTS MADE FROM CRES PH17-4 SHALL NOT BE PROCURED, SELECTED OR SPECIFIED FOR DESIGN AFTER 1 JAN 1983.

12. BOLT PLUNGER MANUFACTURED FROM PH 13-8Mo AND A-286 MATERIAL SHALL BE COATED WITH DRY FILM LUBRICANT PER MIL-L-8937 AND MIL-L-46010 RESPECTIVELY.

13. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BIDS OR REQUEST FOR PROPOSAL, EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.

14. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

15. BOLTS WITH MS33750 RECESS ARE INACTIVE FOR DESIGN AFTER 15 MAY 1981 AND CANCELLED AFTER 1 JUN 1981.

APPROVED 15 DEC 67 REVISED (F) FOR CHANGES SEE SHEET 6

P.A. NAVY - AS. Other Cust: USAF - 11, ARMY - AV	TITLE: BOLTS, SELF-RETAINING, POSITIVE LOCKING, CRES 90 KSI Fsu, 63 KSI Fsu, PAN HEAD, 450°F & 650°F	MILITARY STANDARD MS21125
PROCUREMENT SPECIFICATION: MIL-B-23964	SUPERSEDES: MS21125(ASG)	SHEET 6 OF 6

DD FORM 1 MAR 72 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PLATE NO. 23066

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